Localcoin

Case Study

This take-home assessment is focused on designing a banking system with support for compliance and it is benchmarked to last up to 1 hour long. For that, we`ll focus on the following business domain concepts

- Branch
- Account
- Balance
- Transaction (Transfer, Withdraw, Deposit, Payment)
- Legal Person Identity and Address

Task 1:

Design the relational database tables for the business domain above. For that, use SQL to generate a MySQL-compatible file with all the schema & table definitions. Save your work into **task1.sql**.

Task 2:

Access this link and create a Spring Boot project. Map the database tables into JPA entities, using JPA annotations. Add the controller endpoints:

- POST /api/v1/accounts: Open an account for a legal person at a branch. Body includes identity details (name, DOB etc.), registered address, branch ID, and currency. Returns 201 with the new account ID and status.
- GET /api/v1/accounts/{accountId}: Return account details plus current balance and holder identity. 404 if not found.
- POST /api/v1/accounts/{accountId}/withdrawals: Record a withdrawal

Save your project into a **task2.zip** file.

Submission

Submit all the files directly to Greenhouse no later than 2 days from the time you`ve received this assignment.

Use of AI tools: You may use AI to generate boilerplate, Spring annotations, and other repetitive code blocks. However, you must manually design and write the database schema and business logic. Official documentation and search engines are permitted. Submissions that appear to rely on AI for core design or business logic may be set aside, and we will ask you to walk through your design decisions in detail during the follow-up interview.